LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief description and purpose of inverted flight valve and recuperate
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief description and purpose of inverted flight valve and recuperate.
5. Specific Objectives	23.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Nil
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the inverted flight valve?
(b) Why is fuel flow critical during inverted flight?
(c) What happens if the inverted flight valve fails to close?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Construction of inverted flight valve • Purpose of inverted flight valve • Operation during inverted flight • Recuperate valve function • Recuperate valve operation	L & Ds	PPT, IB	15 min
(b)	• Fuel system requirements for inverted flight • Prevention of fuel starvation • Recuperation process explanation • Valve actuation mechanisms • Safety implications of valve failure	L & Ds	PPT, PDF	20 min
(c)	• Comparison of valve types • Maintenance considerations • Troubleshooting valve issues • Integration with fuel system • Operational limitations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the purpose of the inverted flight valve in preventing fuel starvation.
(b) Describe how the recuperate valve functions during normal flight operations.

12. Home Assignment and Follow Up	Prepare a brief note on how the inverted flight valve and recuperate valve work together to ensure fuel system integrity, referencing the next lesson on pressure refuelling.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principles of pressure refuelling and de-fuelling	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min